

Hierarchical Clustering — Divisive (Top-Down) Guide

Student guide to the top-down form of hierarchical clustering, with practical code for visualizing and cutting a hierarchical tree.

1. Definition

Divisive Hierarchical Clustering is a top-down approach. It starts with all observations in one cluster and repeatedly splits clusters into smaller groups.

2. Agglomerative vs Divisive

Agglomerative	Divisive
Bottom-up	Top-down
Starts with individual points	Starts with one large cluster
Merges clusters	Splits clusters
Merge until one hierarchy remains	Split until desired detail is reached

3. Dataset

Create **customers.csv**:

```
Customer,Annual_Income,Spending_Score
A,20,80
B,22,85
C,25,78
D,60,30
E,62,25
F,65,28
```

4. Divisive concept

Start:

```
[A B C D E F]
```

Split:

```
[A B C] [D E F]
```

Further split:

```
[A B] [C] [D E] [F]
```

Continue until the required level of detail is reached.

Important: standard SciPy's **linkage()** function builds an agglomerative hierarchy. Scikit-learn also does not provide a simple standard DivisiveClustering class. Therefore, the practical code below uses the standard hierarchical dendrogram to teach hierarchy interpretation and cutting; it should not be described as a native divisive implementation.

5. Practical dendrogram code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster

# 1. Read dataset
df = pd.read_csv("customers.csv")
print(df)
```

```

# 2. Select features
X = df[["Annual_Income", "Spending_Score"]]

# 3. Scale data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 4. Create hierarchical linkage
linked = linkage(X_scaled, method="ward")

# 5. Plot dendrogram
plt.figure(figsize=(10, 6))
dendrogram(
    linked,
    labels=df["Customer"].values
)
plt.title("Hierarchical Dendrogram")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()

# 6. Cut the hierarchy using a distance threshold
clusters = fcluster(
    linked,
    t=5,
    criterion="distance"
)

# 7. Add cluster labels
df["Cluster"] = clusters

print("\nFinal Result:")
print(df)

```

6. Deep dive: dendrogram interpretation

The x-axis lists observations. The y-axis shows the distance or linkage height at which groups are joined. A low merge height indicates relatively similar groups; a high merge height indicates greater dissimilarity. The tree lets you inspect the hierarchy at different levels.

7. Cutting the dendrogram

A **height threshold** is a horizontal cut through the dendrogram. The branches separated by that cut form the resulting clusters. For example, if a cut at distance 5 crosses two major branches, the result is two clusters.

```

# Example: choose a height threshold
threshold = 5

clusters = fcluster(
    linked,
    t=threshold,
    criterion="distance"
)

df["Cluster"] = clusters
print(df)

```

8. Choosing the number of clusters

Look for a large vertical gap between successive merge heights. A cut placed inside that gap often gives a useful candidate clustering. This is a heuristic, not a universal guarantee of an optimal solution. Validate the resulting clusters with domain knowledge and, where appropriate, clustering metrics.

9. Comparing linkage effects

```

methods = ["single", "complete", "average", "ward"]

for method in methods:
    linked = linkage(X_scaled, method=method)

```

```
plt.figure(figsize=(10, 5))
dendrogram(
    linked,
    labels=df["Customer"].values
)
plt.title(f"{method.title()} Linkage")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()
```

10. When is divisive useful?

Divisive thinking can be useful when the analysis naturally starts with a broad population and you want to understand progressively finer subgroups. It can be computationally expensive, so agglomerative methods are more common in standard Python workflows.

11. Student exercise

1. Plot a Ward dendrogram.
2. Try thresholds 3, 5, and 7 and observe how the number of clusters changes.
3. Compare Single, Complete, Average, and Ward dendrograms.
4. Explain what dendrogram height means.
5. Explain the difference between top-down divisive and bottom-up agglomerative clustering.